

Math 113 – Abstract Algebra

Chapter-by-chapter notes on the important ideas

Fraleigh, *A First Course in Abstract Algebra* (7th ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Taking notes

Find the chapter banner, then type under the section heading as you read:

`\notesec{1A}{Span and Linear Independence}`

Record big ideas with `idea`, definitions with `definition`, theorems with `theorem`, `lemma`, `proposition`, examples with `example`, and warnings with `remark` — all numbered within the chapter. Add a missing section by copying a `\notesec` slot; add a chapter with `\chapterbanner`.

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Chapter 0 Sets and Relations

Fraleigh: sets, relations, equivalence relations, partitions, functions

§0 Sets and Relations

Chapter 1 Groups and Subgroups

Fraleigh: examples, binary operations, isomorphic binary structures, groups, subgroups, cyclic groups

§1 Introduction and Examples

§2 Binary Operations

§3 Isomorphic Binary Structures

§4 Groups

§5 Subgroups

§6 Cyclic Groups

§7 Generating Sets and Cayley Digraphs

Chapter 2 Permutations, Cosets, and Direct Products

Fraleigh: groups of permutations, orbits and cycles, Lagrange's theorem, direct products, finitely generated abelian groups

§8 Groups of Permutations

§9 Orbits, Cycles, and the Alternating Groups

§10 Cosets and the Theorem of Lagrange

§11 Direct Products and Finitely Generated Abelian Groups

§12 †Plane Isometries

Chapter 3 Homomorphisms and Factor Groups

Fraleigh: homomorphisms, normal subgroups, factor groups, simple groups, group actions

§13 Homomorphisms**§14 Factor Groups****§15 Factor-Group Computations and Simple Groups****§16 ‡Group Action on a Set****§17 †Applications of G -Sets to Counting**

Chapter 4 Rings and Fields

Fraleigh: rings, fields, integral domains, Fermat's and Euler's theorems, quotient fields, polynomial rings and their factorization

§18 Rings and Fields

§19 Integral Domains

§20 Fermat's and Euler's Theorems

§21 The Field of Quotients of an Integral Domain

§22 Rings of Polynomials

§23 Factorization of Polynomials over a Field

§24 †Noncommutative Examples

§25 †Ordered Rings and Fields

Chapter 5 Ideals and Factor Rings

Fraleigh: ring homomorphisms, factor rings, prime and maximal ideals, PIDs

§26 Homomorphisms and Factor Rings

§27 Prime and Maximal Ideals

§28 †Gröbner Bases for Ideals

Chapter 6 Extension Fields

Fraleigh: extension fields, vector spaces, algebraic extensions, finite fields

§29 Introduction to Extension Fields

§30 Vector Spaces

§31 Algebraic Extensions

§32 †Geometric Constructions

§33 Finite Fields

Chapter 7 Advanced Group Theory

Fraleigh: isomorphism theorems, series of groups, the Sylow theorems and applications, free groups, presentations

§34 Isomorphism Theorems

§35 Series of Groups

§36 Sylow Theorems

§37 Applications of the Sylow Theory

§38 Free Abelian Groups

§39 Free Groups

§40 Group Presentations

Chapter 9 Factorization

Fraleigh: unique factorization domains, Euclidean domains, Gaussian integers and multiplicative norms

§45 Unique Factorization Domains

§46 Euclidean Domains

§47 Gaussian Integers and Multiplicative Norms

Chapter 10 Automorphisms and Galois Theory

Fraleigh: automorphisms of fields, the isomorphism extension theorem, splitting fields, separability, the Galois correspondence

§48 Automorphisms of Fields

§49 The Isomorphism Extension Theorem

§50 Splitting Fields

§51 Separable Extensions

§52 †Totally Inseparable Extensions

§53 Galois Theory

§54 Illustrations of Galois Theory

§55 Cyclotomic Extensions

§56 Insolvability of the Quintic
